

Posterior thalamic hemorrhage induces “pusher syndrome”

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Abstract—Background: Recent findings argue for a pathway in humans for sensing the orientation of gravity and controlling upright body posture, separate from the one for orientation perception of the visual world. Stroke patients with contraversive pushing were shown to experience their body as oriented upright when actually tilted about 20° to the ipsilesional side, in spite of normal visual-vestibular functioning. A recent study suggested the involvement of posterolateral thalamus typically associated with the disorder. **Objective:** To evaluate the relationship between pushing behavior and thalamic function. **Methods:** Over a 3-year period the authors prospectively investigated 40 patients with left- or right-sided thalamic strokes. **Results:** Twenty-eight percent showed contraversive pushing. The authors found a strong relationship between etiology, vascular territory, lesion size, and neurologic disorders associated with contraversive pushing. Pusher patients had larger lesions that typically were caused by hemorrhage (vs infarcts) located in the posterior thalamus (vs anterior thalamic lesions in those patients without pushing behavior). A paresis of the contralesional extremities was more frequent and more severe in pusher patients. Further, these patients showed more additional spatial neglect with right thalamic lesions, while they tended to be more aphasic with left thalamic lesions. **Conclusions:** Posterior thalamus seems to be fundamentally involved in our control of upright body posture. Higher pressure, swelling, and other secondary pathologic processes associated with posterior thalamic hemorrhage (vs thalamic infarction) may provoke contraversive pushing in combination with additional neurologic symptoms.

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Hemiparetic stroke patients with “pusher syndrome” show the peculiar behavior of using the nonaffected arm or leg to push away actively from the nonparalyzed side.^{1,2} Without assistance, this active contraversive pushing leads to a loss of postural balance and falling toward the paralyzed side. Nevertheless, the patients resist any attempt to correct their tilted body posture toward the earth-vertical upright orientation. The disorder is observed in about 10% of acute stroke patients with hemiparesis.³

Patients with contraversive pushing experience their body as oriented upright when actually tilted about 20° to the ipsilesional side.⁴ Surprisingly, these patients show undisturbed processing of visual and vestibular inputs determining visual vertical. Thus, in contrast to their disturbed perception of upright body posture, orientation perception of the visual world is unaffected.⁴ This finding argued for a pathway in humans for sensing the orientation of gravity and controlling upright body posture, separate from the one for orientation perception of the visual world.⁴

A prior study could not identify the neural correlates of contraversive pushing, probably due to the low spatial resolution of the method used for lesion analysis.³ It divided the entire brain into only eight large sections and determined the frequency of their

involvement in the individual CT lesions. The authors found a difference between patients with and without contraversive pushing only for the white matter in the internal capsule. New tools became available that allowed more precise lesion localization in humans.⁵ Using such techniques, a sample of 46 consecutively admitted patients with and without contraversive pushing with unselected cortical or subcortical lesions was investigated.⁶ The analysis revealed that the brain structure typically damaged in left as well as in right brain damaged pusher patients is the posterolateral thalamus. Thus, the present study evaluated the relationship between contraversive pushing and thalamic function. Over a 3-year period we prospectively investigated all patients with left- or with right-sided thalamic strokes admitted to our department.

Methods. Forty patients with thalamic strokes consecutively admitted from a well-defined recruitment area belonging to the University Hospital in Tübingen were included. The lesions were demonstrated by MRI or by CT. Patients with diffuse or bilateral brain lesions or patients with tumors were excluded. Following a standardized testing for pushing behavior (see below), the 40 patients were divided into groups with and without contraversive pushing (table). All patients gave written informed consent to participate in the study, which was in accordance with the ethical standards of the 1964 Declaration of Helsinki.

Clinical investigation. Demographic and clinical data of all subjects are presented in table 1. Contraversive pushing was as-

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Table Demographic and clinical data of the 40 patients with left or right thalamic lesions showing or not showing pushing behavior

	R Pusher	R Controls	L Pusher	L Controls
Number	9	14	5	12
Sex, F/M	4/5	5/9	2/3	6/6
Age, y, mean (SD)	66.1 (7.5)	62.2 (12.2)	63.9 (9.7)	61.8 (18.0)
Etiology, infarct/hemorrhage	2/7	11/3	0/5	9/3
Lesion volume, % of hemisphere, mean (SD)	0.6 (0.4)	0.3 (0.2)	1.0 (0.6)	0.3 (0.2)
Time since lesion—clinical exam, d, mean (SD)	6.2 (2.5)	4.7 (4.1)	9.4 (3.8)	3.8 (2.9)
SCP Posture, median (range)				
Sitting	1 (0.25–1)	0	1 (0.25–1)	0
Standing	1 (0.75–1)	0	1	0
SCP Extension, median (range)				
Sitting	0.5 (0.5–1)	0	1 (0–1)	0
Standing	1 (0.5–1)	0	1	0
SCP Resistance, median (range)				
Sitting	1 (0–1)	0	1 (0–1)	0
Standing	1	0	1	0
Visual field defect, % present	0	0	0	0
Paresis of contralesional side, % present	100	43	100	50
Arm, median (range)	2 (0–4.5)	5 (0–5)	1 (0–2)	4.5 (1–5)
Leg, median (range)	3.5 (1–4.5)	5 (2.5–5)	1 (0–4)	5 (3–5)
Somatosensory deficit of contralesional side (touch), % present	89	43	80	58
% TNP	11	0	0	0
Aphasia, % present	0	0	60	25
Spatial neglect, % present	67	21	0	8

SCP = Scale for Contraversive Pushing^{2,4}; TNP = testing not possible.

essed by using the standardized Scale for Contraversive Pushing (SCP).^{2,4} The SCP assesses 1) symmetry of spontaneous posture, 2) the use of the nonparetic arm or leg to increase pushing force by abduction and extension of extremities, and 3) resistance to passive correction of posture. Pusher behavior was diagnosed when the patients showed at least a total score of 1 (maximum = 2, sitting plus standing) with respect to their spontaneous posture, at least a score of 1 (maximum = 2, sitting plus standing) concerning the use of the nonparetic arm or leg to increase pushing force by abduction and extension, and when they showed resistance to passive correction of posture. (In five patients, pushing behavior while standing could not be quantified with SCP due to a complete inability to reach a standing position at the time of the present investigation.)

Visual field defects were assessed by standardized neurologic examination. The degree of paresis of the upper and lower limbs was scored with the usual clinical ordinal scale, where 0 stands for no trace of movement and 5 for normal movement. Spatial neglect was diagnosed when the patients showed the characteristic clinical behavior such as orienting toward the ipsilesional side when addressed from the front or the left or ignoring of contralesionally located people or objects. In addition, all patients were further assessed with the following four clinical tests: the Letter cancellation task,⁷ the Bells test,⁸ the Baking tray task,⁹ and a copying task.¹⁰ Neglect patients had to fulfill the criterion for spatial neglect in at least two of these four clinical tests.¹¹ Aphasia was assessed using a bedside examination that evaluated spontaneous speech, auditory and reading comprehension, picture naming, reading, and oral repetition.

Lesion analysis. Using two standard protocols, MRI was carried out in 14 of the patients with stroke and CT imaging in 26 patients. Under both protocols the initial scanning optionally was repeated during the following days until a firm diagnosis could be

made and the affected area became clearly demarcated. The final scans were used for the present study.

In those patients who underwent the MRI protocol, we used diffusion-weighted imaging (DWI) within the first 48 hours post-stroke and T2-weighted fluid-attenuated inversion recovery (FLAIR) sequences when imaging was conducted 48 hours or later after the stroke. While FLAIR images provide high sensitivity for acute cerebral infarcts, DWI has proved to be particularly sensitive for the detection of hyperacute infarcts and shows high accuracy in predicting final infarct size.^{12–15} Scans were obtained on a 1.5-T echoplanar imaging (EPI) capable system (Magnetom Sonata, Siemens, Erlangen, Germany). The FLAIR sequence was acquired with 20 axial slices (thickness 5 mm) with an interslice gap of 1 mm, a field of view (FOV) of 175 × 220 mm², a repetition time (TR) of 9,000 msec, and an echo time (TE) of 118 msec. DWI was performed with a single-shot EPI spin echo sequence (TR 3,200 msec, TE 87 msec, FOV 230 × 230 mm², matrix 64 × 64 pixels, slice thickness 5 mm; gap 1 mm). The mean time between the onset of stroke and the MRI used for the present study was 7.5 days (SD 15.7). The boundary of the lesion was delineated directly on the individual MRI for every single transversal slice using MRIcro software¹⁶ (<http://www.mricro.com>). Both the scan and lesion shape were then transferred into stereotaxic space using the spatial normalization algorithm provided by SPM2 (<http://www.fil.ion.ucl.ac.uk/spm/>). For determination of the transformation parameters, cost-function masking was employed.¹⁷

To fit approximately the canonical AC-PC orientation of the MR scans, the CT imaging protocol used the line drawn between the occiput and the lower margin of the orbita to orient the scans in each individual. Scans were obtained on a spiral CT scanning system (Somatom Sensation 16, Siemens, Erlangen, Germany) that has created excitement because it provides clearer pictures with more anatomic details and in less time than previous gener-

ations of CT systems.⁵ Spiral CT scanning was performed with a slice thickness of 3 mm infratentorial and 8 mm supratentorial. MRIcro software was used to map the lesion on transversal slices of the T1-template MRI from the Montreal Neurologic Institute (www.bic.mni.mcgill.ca/cgi/icbm_view) that likewise is aligned with stereotaxic space and is distributed with MRIcro. The template scan provides various anatomic landmarks for precisely plotting size and localization of the lesion. The mean time between the onset of stroke and the CT used for the present study was 8.0 days (SD 10.7) when the affected area became clearly demarcated. Lesions were mapped onto the slices that correspond to Talairach Z-coordinates -40, -32, -24, -16, -8, 0, 8, 16, 24, 32, 40, and 50 mm in Talairach space by using the identical or the closest matching transversal slices of each individual.

Lesion location in thalamic stroke patients with and without contraversive pushing was compared using the subtraction technique.⁵ This analysis illustrates the center of overlap in the patients with pusher behavior in direct visual contrast to those areas that do not induce the disorder when lesioned. The resulting subtraction image specifically highlights regions that are both frequently damaged in patients with contraversive pushing as well as being typically spared in patients without that disorder. Because the two patient groups differed in sample size, we used proportional values for the MRIcro subtraction analysis.

Results. In the 3-year sample of 40 patients with thalamic strokes, we found contraversive pushing in 28% of the subjects. We compared the two groups with left- and right-sided strokes who demonstrated pushing behavior with the two groups with left- and right-sided thalamic damage who did not show the disorder.

Clinical differences between thalamic patients with and without contraversive pushing. We analyzed age, etiology of stroke, frequency of additional spatial neglect, aphasia, and visual field defects, strength of contralateral paresis, and frequency of contralateral sensory loss between the patients with and without contraversive pushing. In left as well as in right brain damaged subjects, no differences in age were observed between pusher patients and those who did not show contraversive pushing (Student's *t*-test for independent samples: both $t < 0.837$, both $p > 0.412$).

In patients with right-sided thalamic lesions, we found a higher frequency of spatial neglect in pusher patients than in those without contraversive pushing (χ^2 test: $\chi^2 = 4.71$, $p < 0.05$). In patients with left-sided thalamic lesions, the frequency of aphasia was higher in the patients showing contraversive pushing than in those without but did not reach significance ($\chi^2 = 1.89$, $p > 0.05$). Moreover, contralesional paresis was more frequent (right-sided lesions: $\chi^2 = 7.89$, left-sided lesions: $\chi^2 = 3.86$, both $p < 0.05$) and was more severe in pusher patients for both arm and leg (Mann-Whitney *U* test: right-sided: both $U > 16.5$, $p < 0.007$; left-sided: both $U > 2.0$, $p < 0.003$). The numerically higher frequency of somatosensory loss in left-sided as well as in right-sided pusher patients did not reach significance (both $\chi^2 < 3.65$, both $p > 0.05$). None of the subjects showed visual field defects.

The analysis of stroke etiology revealed that 86% of the patients with pushing behavior had thalamic hemorrhage while only 23% showed this etiology in the group of patients without the disorder ($\chi^2 = 14.43$, $p < 0.05$).

Anatomic differences between thalamic patients with and without contraversive pushing. Lesion size was larger in the left brain damaged pusher patients (Mann-Whitney *U* test: $U = 4.5$, $p = 0.006$) and tended to be larger in right brain damaged pusher patients ($U = 32.5$, $p = 0.052$), vs the patient groups without contraversive pushing. Figure 1 illustrates overlay plots for the group of

pusher patients with left-sided lesions (L-pusher) and with right-sided lesions (R-pusher). The number of overlapping lesions is illustrated by different colors coding increasing frequencies from violet ($n = 1$) to red ($n = \text{maximum number of subjects in the respective group}$). Similar plots are shown for the two control groups. To identify the thalamic structures that are commonly damaged in patients with contraversive pushing but are typically spared in patients with left or with right thalamic lesions but no pushing behavior, we subtracted the superimposed lesions of the control groups from the overlap images of the groups with contraversive pushing revealing percentage overlay plots (figure 2). This method creates an image that shows regions that are commonly damaged in pusher patients but are typically spared in control patients (coded as positive values). Likewise, it shows regions specifically damaged in the control patients but typically spared in pusher patients (coded as negative values). In figure 2, we used progressively brighter shades of orange to highlight positive values and progressively brighter shades of blue to illustrate negative values.

We found a clear separation of lesion overlap between pusher patients and controls. In both hemispheres, the anatomic structures that were damaged more frequently in the patient groups with contraversive pushing than in controls centered on the posterior thalamus (see figure 2). The area of greatest lesion overlap extended in the left hemisphere from Talairach coordinates ($x, -14; y, -30; z, 0$) over ($x, -14; y, -30; z, 8$) to ($x, -12; y, -24; z, 16$) and in the right hemisphere from Talairach coordinates ($x, 14; y, -25; z, 0$) over ($x, 12; y, -27; z, 8$) to ($x, 16; y, -22; z, 16$). In contrast, regions specifically affected more frequently in the control patients than in pusher patients (coded as blue values in figure 2) centered on the anterior thalamus.

Discussion. Evaluating a consecutively admitted 3-year sample of 40 patients with thalamic strokes, we found contraversive pushing in 28% of subjects. These patients showed a typical pattern of clinical and anatomic pathology that clearly differed from thalamic patients without pusher behavior. We found that pusher patients had larger lesions that typically were caused by thalamic hemorrhage located in the posterior thalamus.

This result seems to resemble a natural relationship between the two stroke etiologies (hemorrhage vs infarct) on the one hand and the vascular territories affected within the thalamus on the other. It is known that 1) thalamic hemorrhages predominantly affect the posterolateral part of the thalamus¹⁸⁻²¹ and that 2) infarctions are less frequent in the posterior thalamus vs the anterior and paramedian thalamus.^{22,23} This relationship between etiology and vascular territory still is not well understood. It might be related to the fact that the thalamogeniculate arteries are the largest and the most consistent of the arterial branches penetrating the thalamus.²¹ The present observation that contraversive pushing typically is due to hemorrhage (vs infarcts) and typically associated with a lesion of the posterior thalamus (vs anterior thalamic lesions in those patients without

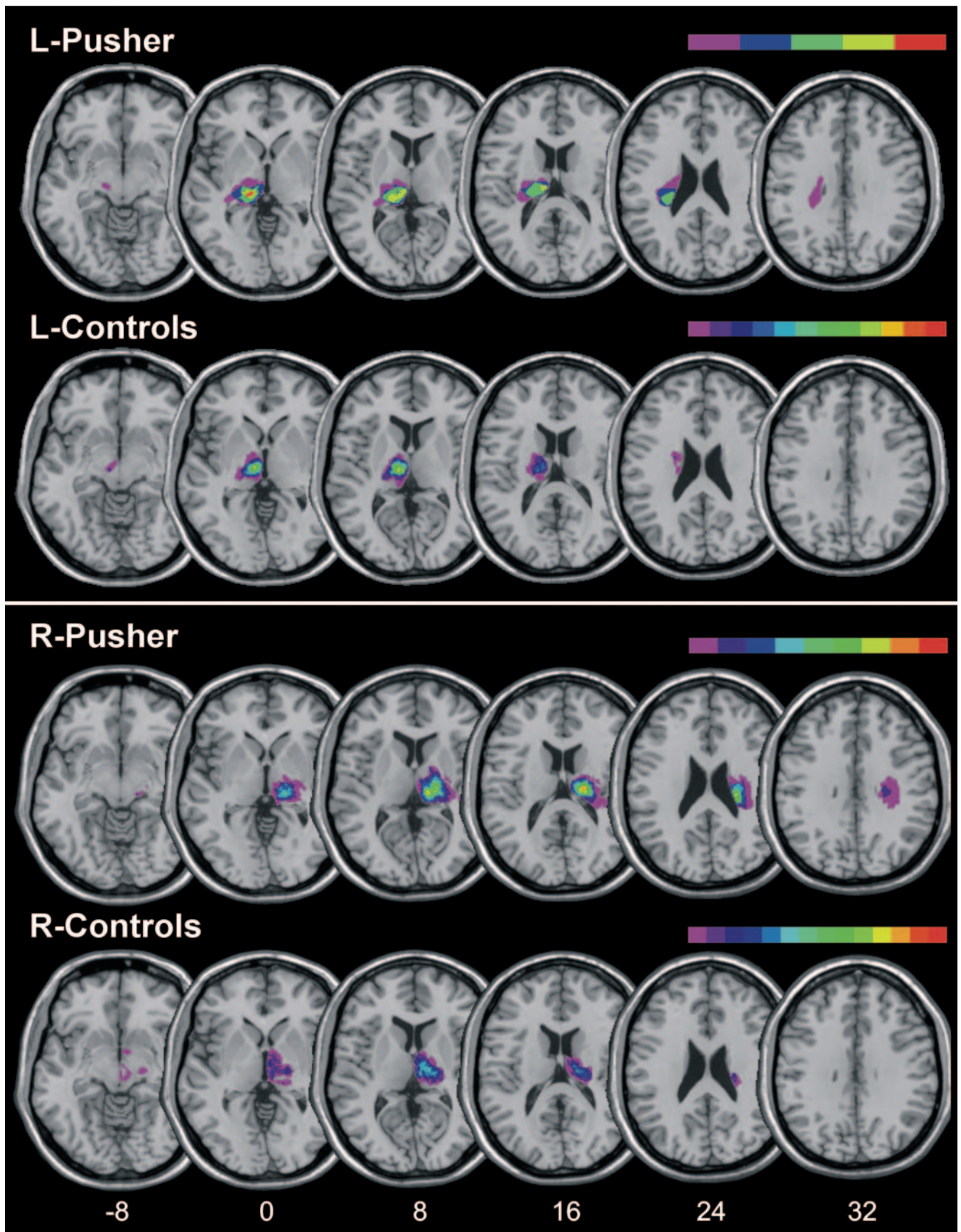


Figure 1. Overlay lesion plots of the patients with contraversive pushing ($n = 5$ with left-sided lesions [L], $n = 9$ with right-sided lesions) and control patients ($n = 12$ with left-sided lesions [L], $n = 14$ with right-sided lesions). The number of overlapping lesions is illustrated by different colors coding increasing frequencies from violet ($n = 1$) to red ($n = \text{maximum number of subjects in the respective group}$). Talairach z-coordinates⁴² of the transverse sections are given.

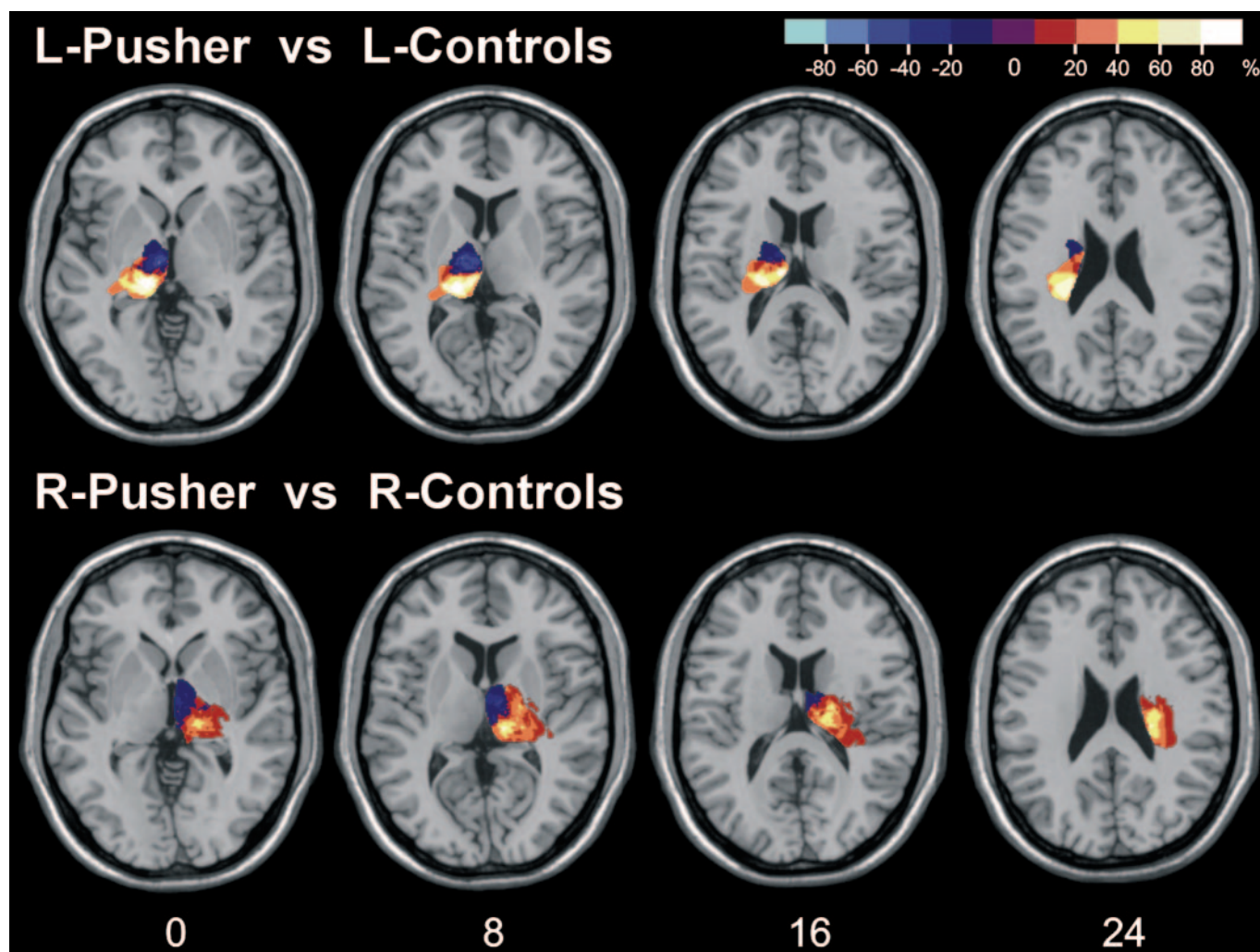


Figure 2. Overlay plots of the subtracted superimposed lesions of the left (L) and of the right (R) brain damaged groups with contraversive pushing minus their respective controls. The percentage of overlapping lesions of the pusher patients after subtraction of controls is illustrated by five different colors coding increasing frequencies from dark red (difference = 1% to 20%) to white (difference = 81% to 100%). Each color represents 20% increments. The different colors from dark blue (difference = -1% to -20%) to light blue (difference = -81% to -100%) indicate regions damaged more frequently in control patients than in pusher patients. Talairach z-coordinates⁴² of each transverse section are given.

pushing behavior) thus are well consistent with these observations.

The present study also revealed that a paresis of the contralesional extremities was more frequent and more severe in pusher patients than in patients without the disorder. Further, patients with contraversive pushing showed more additional spatial neglect with right thalamic lesions, while they tended to be more aphasic with left thalamic lesions. Our study cannot differentiate between the distinct influence of lesion location, size of lesion, and etiology of the lesion in the genesis of contraversive pushing and additional neurologic disorders. We can only speculate that higher pressure and other secondary pathologic processes associated with posterior thalamic hemorrhage (vs thalamic infarction) might cause this typical combination of disorders. The local edema associated with thalamic infarction may cause rather limited increase in tissue pressure,

while the presence of a hematoma within the thalamus probably put larger irritation, swelling, and pressure on thalamic as well as surrounding tissues. For example, more extensive affection of the adjacent internal capsule in the case of thalamic hemorrhage could well explain the more severe paresis observed in thalamic patients with contraversive pushing compared to those without.

Our findings support that the posterior thalamus is fundamentally involved in the control of upright body posture. From neurophysiologic work in monkeys, we know that some nuclei in this portion of the thalamus are sensitive to vestibular stimulation.^{24,25} These nuclei therefore had been regarded as vestibular relay structures to the cortex.²⁶ In line with this idea was the observation that 69% of patients with acute infarctions of this area showed a tilt of the subjective visual vertical.²⁷ Our present and recent data^{4,6} teach us that the posterior thalamus is more

than a relay structure of the vestibular pathway. Rather it may be fundamentally involved in the neural representation of a second graviceptive system in humans decisive for our control of upright body posture. It was discovered that pusher patients show an altered perception of the body's orientation in relation to gravity, while processing of visual and vestibular inputs determining visual vertical is undisturbed.⁴ Experiments in patients with acute unilateral vestibular lesions reported the opposite dissociation.^{28,29} Whereas these latter patients showed a tilted perception of visual vertical, their perception of own body orientation with reference to gravity remained veridical. This double dissociation argued for different pathways subserving these two perceptions. It seems that a separate pathway exists in humans for sensing the orientation of gravity apart from the well known pathway for orientation perception of the visual world.

Anatomic studies in nonhuman primates^{30–38} as well as recent diffusion tensor MRI studies in living human subjects^{39–41} have demonstrated the rich connections between the posterior thalamus and occipital, parietal, and temporal cortical structures. The question thus arises whether lesion or metabolic abnormalities in these cortical areas likewise are related to pusher behavior. Future studies have to investigate whether—in addition to the posterior thalamus—(part of) these cortical structures contribute to our control of upright body posture.

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References

- Davies PM. Steps to follow. A guide to the treatment of adult hemiplegia. New York: Springer, 2000.
- Karnath H-O, Broetz D. Understanding and treating "pusher syndrome." *Phys Ther* 2003;83:1119–1125.
- Pedersen PM, Wandel A, Jorgensen HS, Nakayama H, Raaschou HO, Olsen TS. Ipsilateral pushing in stroke: incidence, relation to neuropsychological symptoms, and impact on rehabilitation. The Copenhagen Stroke Study. *Arch Phys Med Rehabil* 1996;77:25–28.
- Karnath H-O, Ferber S, Dichgans J. The origin of contraversive pushing: evidence for a second graviceptive system in humans. *Neurology* 2000;55:1298–1304.
- Rorden C, Karnath H-O. Using human brain lesions to infer function: a relic from a past era in the fMRI age? *Nat Rev Neurosci* 2004;5:813–819.
- Karnath H-O, Ferber S, Dichgans J. The neural representation of postural control in humans. *Proc Natl Acad Sci USA* 2000;97:13931–13936.
- Weintraub S, Mesulam M-M. Mental state assessment of young and elderly adults in behavioral neurology. In: Mesulam M-M, ed. *Principles of behavioral neurology*. Philadelphia: F.A. Davis, 1985;71–123.
- Gauthier L, Dehaut F, Joanette Y. The bells test: a quantitative and qualitative test for visual neglect. *Int J Clin Neuropsychol* 1989;11:49–54.
- Tham K, Tegnér R. The Baking Tray Task: A test of spatial neglect. *Neuropsychol Rehabil* 1996;6:19–25.
- Johannsen L, Karnath H-O. How efficient is a simple copying task to diagnose spatial neglect in its chronic phase? *J Clin Exp Neuropsychol* 2004;26:251–256.
- Karnath H-O, Himmelbach M, Rorden C. The subcortical anatomy of human spatial neglect: putamen, caudate nucleus, and pulvinar. *Brain* 2002;125:350–360.
- Brant-Zawadzki M, Atkinson D, Detrick M, Bradley WG, Scidmore G. Fluid-attenuated inversion-recovery (FLAIR) for assessment of cerebral infarction: initial clinical experience in 50 patients. *Stroke* 1996;27:1187–1191.
- Noguchi K, Ogawa T, Inugami A, et al. MRI of acute cerebral infarction: a comparison of FLAIR and T2-weighted fast spin-echo imaging. *Neuroradiol* 1997;39:406–410.
- Ricci PE, Burdette JH, Elster AD, Reboussin DM. A comparison of fast spin-echo, fluid-attenuated inversion-recovery, and diffusion-weighted MR imaging in the first 10 days after cerebral infarction. *Am J Neuroradiol* 1999;20:1535–1542.
- Schaefer PW, Hunter GJ, He J, et al. Predicting cerebral ischemic infarct volume with diffusion and perfusion MR imaging. *Am J Neuroradiol* 2002;23:1785–1794.
- Rorden C, Brett M. Stereotaxic display of brain lesions. *Behav Neurol* 2000;12:191–200.
- Brett M, Leff AP, Rorden C, Ashburner J. Spatial normalization of brain images with focal lesions using cost function masking. *NeuroImage* 2001;14:486–500.
- Hungerbühler JP, Assal G, Regli F. Thalamic hematomas: neuropsychological aspects. Report of 11 cases and review of literature. *Schweiz Arch Neurol Neurochir Psychiatr* 1984;135:199–215.
- Kawahara N, Sato K, Muraki M, Tanaka K, Kaneko M, Uemura K. CT classification of small thalamic hemorrhages and their clinical implications. *Neurology* 1986;36:165–172.
- Kumral E, Kocaer T, Ertubey NO, Kumral K. Thalamic hemorrhage. A prospective study of 100 patients. *Stroke* 1995;26:964–970.
- Chung CS, Caplan LR, Han W, Pessin MS, Lee KH, Kim JM. Thalamic haemorrhage. *Brain* 1996;119:1873–1886.
- Bogousslavsky J, Regli F, Uske A. Thalamic infarcts: clinical syndromes, etiology, and prognosis. *Neurology* 1988;38:837–848.
- Van der Werf YD, Witter MP, Uylings HBM, Jolles J. Neuropsychology of infarctions in the thalamus: a review. *Neuropsychologia* 2000;38:613–627.
- Deeke L, Schwarz DWF, Fredrickson JM. Nucleus ventroposterior inferior (VPI) as the vestibular thalamic relay in the rhesus monkey. I. Field potential investigation. *Exp Brain Res* 1974;20:88–100.
- Büttner U, Henn V. Thalamic unit activity in the alert monkey during natural vestibular stimulation. *Brain Res* 1976;103:127–132.
- Brandt T, Dieterich M, Danek A. Vestibular cortex lesions affect the perception of verticality. *Ann Neurol* 1994;35:403–412.
- Dieterich M, Brandt T. Thalamic infarctions: differential effects on vestibular function in the roll plane (35 patients). *Neurology* 1993;43:1732–1740.
- Bisdorff AR, Wolsley CJ, Anastasopoulos D, Bronstein AM, Gresty MA. The perception of body verticality (subjective postural vertical) in peripheral and central vestibular disorders. *Brain* 1996;119:1523–1534.
- Anastasopoulos D, Haslwanter T, Bronstein A, Fetter M, Dichgans J. Dissociation between the performance of body verticality and the visual vertical in acute peripheral vestibular disorder in humans. *Neurosci Lett* 1997;233:151–153.
- Pearson RC, Brodal P, Powell TP. The projection of the thalamus upon the parietal lobe in the monkey. *Brain Res* 1978;144:143–148.
- Jones EG. *The thalamus*. New York, NY: Plenum Press, 1985.
- Yeterian EH, Pandya DN. Thalamic connections of the cortex of the superior temporal sulcus in the rhesus monkey. *J Comp Neurol* 1989;282:80–97.
- Schmahmann JD, Pandya DN. Anatomical investigation of projections from thalamus to posterior parietal cortex in the rhesus monkey: a WGA-HRP and fluorescent tracer study. *J Comp Neurol* 1990;295:299–326.
- Baleydier C, Morel A. Segregated thalamocortical pathways to inferior parietal and inferotemporal cortex in macaque monkey. *Vis Neurosci* 1992;8:391–405.
- Hardy SG, Lynch JC. The spatial distribution of pulvinar neurons that project to two subregions of the inferior parietal lobule in the macaque. *Cereb Cortex* 1992;2:217–230.
- Webster MJ, Bachevalier J, Ungerleider LG. Subcortical connections of inferior temporal areas TE and TEO in macaque monkeys. *J Comp Neurol* 1993;335:73–91.
- Rockland KS, Andresen J, Cowie RJ, Robinson DL. Single axon analysis of pulvinocortical connections to several visual areas in the macaque. *J Comp Neurol* 1999;406:221–250.
- Shipp S. The functional logic of cortico-pulvinar connections. *Phil Trans R Soc Lond B* 2003;358:1605–1624.
- Behrens TEJ, Johansen-Berg H, Woolrich MW, et al. Non-invasive mapping of connections between human thalamus and cortex using diffusion imaging. *Nat Neurosci* 2003;6:750–757.
- Wiegell MR, Tuch DS, Larsson HB, Wedeen VJ. Automatic segmentation of thalamic nuclei from diffusion tensor magnetic resonance imaging. *Neuroimage* 2003;19:391–401.
- Johansen-Berg H, Behrens TEJ, Sillery E, et al. Functional-anatomical validation and individual variation of diffusion tractography-based segmentation of the human thalamus. *Cereb Cortex* 2005;15:31–39.
- Talairach J, Tournoux P. *Coplanar stereotaxic atlas of the human brain: 3-dimensional proportional system—an approach to cerebral imaging*. New York: Thieme, 1988.

14. Poser CM, Paty DW, Scheinberg L, et al. New diagnostic criteria for multiple sclerosis: guidelines for research protocols. *Ann Neurol* 1983; 13:227-231.
15. Kurtzke JF. Rating neurologic impairment in multiple sclerosis: an expanded disability scale. *Neurology* 1983;33:1444-1452.
16. Rao SM. Cognitive Function Study Group. National Multiple Sclerosis Society. A manual for the brief repeatable battery of neuropsychological tests in multiple sclerosis. New York: National Multiple Sclerosis Society, 1990.
17. Zigmond AS, Snaith RP. The Hospital Anxiety and Depression Scale. *Acta Psychiatr Scand* 1983;67:361-370.
18. Sharrack B, Hughes RAC. Guy's Neurological Disability Scale (GNDS): a new disability measure for multiple sclerosis. *Mult Scler* 1999;5:223-233.
19. Rowbotham M, Harden N, Stacey B, Bernstein P, Magnus Miller L. Gabapentin for the treatment of postherpetic neuralgia. *JAMA* 1998; 280:1837-1842.
20. Backonja M, Beydoun A, Edwards KR, et al. Gabapentin for the symptomatic treatment of painful neuropathy in patients with diabetes mellitus. *JAMA* 1998;280:1831-1836.
21. Svendsen KB, Jensen TS, Bach FW. Does the cannabinoid dronabinol reduce central pain in multiple sclerosis? Randomised double-blind placebo controlled crossover trial. *BMJ* 2004;329:253-258.
22. Attal N, Bouhassira D. Can pain be more or less neuropathic? *Pain* 2004;110:510-511.
23. Rasmussen PV, Sindrup SH, Jensen TS, Bach FW. Symptoms and signs in patients with suspected neuropathic pain. *Pain* 2004;110:461-469.
24. Spissu A, Cannas A, Ferrigno P, Pelaghi AE, Spissu M. Anatomic correlates of painful tonic spasms in multiple sclerosis. *Mov Disord* 1999;14:331-335.
25. Osterman PO, Westerberg CE. Paroxysmal attacks in multiple sclerosis. *Brain* 1975;98:189-202.
26. Boureau F, Legallicier P, Kabir-Ahmadi M. Tramadol in post-herpetic neuralgia: a randomized, placebo-controlled trial. *Pain* 2003;104:323-331.
27. Lesser H, Sharma U, LaMoreaux L, Poole RM. Pregabalin relieves symptoms of painful diabetic neuropathy. A randomized controlled trial. *Neurology* 2004;63:2104-2110.
28. Farrar JT, Young JP, LaMoreaux L, Werth JL, Poole RM. Clinical importance of changes in chronic pain intensity measured on an 11-point numerical pain rating scale. *Pain* 2001;94:149-158.
29. Sindrup SH, Jensen TS. Efficacy of pharmacological treatments of neuropathic pain: an update and effect related to mechanism of drug action. *Pain* 1999;83:389-400.
30. Houtchens MK, Richert JR, Sami A, Rose JW. Open-label gabapentin treatment for pain in multiple sclerosis. *Mult Scler* 1997;3:250-253.
31. Ramsaransing G, Zwanikken C, De Keyser J. Drug points: worsening of symptoms of multiple sclerosis associated with carbamazepine. *BMJ* 2000;320:1112.
32. Killestein J, Hoogervorst EL, Reif M, et al. Safety, tolerability and efficacy of orally administered cannabinoids in MS. *Neurology* 2002;58: 1404-1437.
33. Zajicek J, Fox P, Sanders H, et al. Cannabinoids for treatment of spasticity and other symptoms related to multiple sclerosis (CAMS study): multicentre randomised placebo-controlled trial. *Lancet* 2003; 362:1517-1526.
34. Solowij N, Stephens RS, Roffman RA, et al. Cognitive functioning of long-term heavy cannabis users seeking treatment. *JAMA* 2002;287: 1123-1131.
35. Langdon DW, Thompson AJ, Johnson KP, et al. The psychological effects of cannabis in MS: impact on cognition, pain, mood and fatigue. *Mult Scler ECTRIMS* 2003;9:S27. Abstract.

Corrections

Posterior thalamic hemorrhage induces “pusher syndrome”

In the article “Posterior thalamic hemorrhage induces ‘pusher’” syndrome (*Neurology* 2005;64:1014-1019) by Karnath et al., the observed overall percentage of patients with pushing behavior following a thalamic stroke has been erroneously stated as 28%. The correct percentage is 35%, which should have been stated on page 1016, left column, paragraph 1 of Results and on the same page, right column, paragraph 1 of Discussion. The authors apologize for this error.

Familial hemiplegic migraine: More than just a headache

In the editorial “Familial hemiplegic migraine: More than just a headache” (*Neurology* 2005;64:592-593) by Benatar and Ford, Dr. Ford's middle initial was incorrect. The author's correct name is Corey C. Ford.